Business Requirement Document

HRMS Portal with Mobile App, iOS App, Admin Web App, Payroll, Attendance, Recruitment, AI Interview & Workforce Management

1. Project Objective

The objective of this project is to develop a comprehensive **Human Resource Management System** that enables organizations to manage the complete employee lifecycle from hiring to exit through a centralized, secure, and scalable platform.

The proposed HRMS will include:

- Android mobile application
- iOS mobile application
- Admin web application
- Employee self-service portal
- Manager self-service portal
- Payroll and statutory compliance management
- Attendance management using QR, RFID, thumb impression, face/location-based verification
- Leave, shift, holiday, overtime, and regularization workflows
- AI-based recruitment and interview assessment
- Performance, OKR, appraisal, and employee engagement modules
- Expense, claims, helpdesk, timesheet, project, and analytics modules

The system should reduce manual HR operations, improve payroll accuracy, automate attendance and compliance workflows, improve hiring quality through AI, and provide management with real-time workforce insights.

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2. Product Vision

The HRMS portal should work as a **single digital HR operating system** for companies of different sizes.

The platform should be capable of managing:

- Employee master data
- Attendance and leave
- Payroll and salary processing
- Statutory compliance
- Recruitment and onboarding
- AI-based candidate screening and interviewing
- Performance management
- Expense and reimbursement
- Timesheets and project billing
- HR helpdesk
- Reports and analytics

The long-term vision is to make the platform comparable to enterprise HRMS products like Keka, while adding stronger AI-based recruitment, attendance flexibility, and customizable workflows.

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3. Platforms Required

3.1 Admin Web App

Used by HR, payroll team, finance team, admins, managers, and super admins.

Main purposes:

- Configure company policies
- Manage employees
- Run payroll
- Approve leaves, expenses, attendance, interviews
- View dashboards and reports
- Configure roles and permissions
- Manage organization structure
- Monitor attendance devices and integrations

3.2 Android Mobile App

Used by employees, managers, field staff, candidates, and interviewers.

Main purposes:

- Mark attendance
- Apply leave
- View payslip
- Submit expenses
- Attend AI interviews
- View tasks and approvals
- Receive notifications
- Access employee profile and documents

3.3 iOS Mobile App

Same functionality as Android app, optimized for iPhone users.

3.4 Candidate Portal

Can be part of web/mobile or a separate lightweight portal.

Main purposes:

- Candidate registration
- Resume upload
- Job application
- AI interview
- Interview status tracking
- Offer acceptance
- Document submission

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4. Key User Roles

Role	Description
Super Admin	Platform-level admin for managing multiple companies/tenants
Company Admin	Admin for one organization
HR Admin	Manages employee data, onboarding, exits, letters, policies
Payroll Admin	Manages salary, payroll, compliance, payslips
Finance Admin	Handles reimbursements, expenses, payment reports
Manager	Approves attendance, leave, expenses, performance reviews
Employee	Uses ESS/mobile app for HR self-service
Recruiter	Manages jobs, candidates, interviews, hiring pipeline
Interviewer	Conducts manual interviews and reviews AI interview reports
Candidate	Applies for jobs and attends AI interview
Auditor	Read-only access to logs, payroll, compliance, and reports
IT Admin	Manages access, devices, integrations, MFA, and security settings

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5. Module-Wise BRD

Module 1: Organization & Tenant Management

Objective

To allow the platform to support one or multiple organizations with separate employee data, policies, payroll rules, attendance rules, and access controls.

Functional Requirements

Admin Web App

The system should allow the admin to:

- Create company/tenant profile
- Add company logo, address, GST, PAN, TAN, PF, ESI and other registration details
- Configure branches, offices, plants, warehouses, departments, designations, grades, cost centers, and business units
- Define reporting hierarchy
- Define company calendar and working days
- Configure employee code generation
- Configure multi-location policies
- Configure role-based access

Backend Requirements

The backend should support:

- Multi-tenant data separation
- Tenant-wise configuration
- Branch/location-wise policies
- Department and role hierarchy
- Audit logs for all admin changes

Key Entities

- Tenant
- CompanyProfile
- Branch
- Department
- Designation
- Grade
- CostCenter
- BusinessUnit
- Role
- Permission
- UserRoleMapping

Module 2: Employee Information Management

Objective

To maintain a centralized digital employee database.

Functional Requirements

Employee Profile

The system should store:

- Employee code
- Name
- Date of birth
- Gender
- Contact details
- Emergency contact
- Address
- Joining date
- Employment type
- Department
- Designation
- Reporting manager
- Work location
- Grade
- Bank details
- PAN
- Aadhaar
- UAN
- ESIC number
- Tax details
- Salary structure
- Documents
- Experience details
- Education details
- Family details
- Nominee details

Employee Lifecycle

The system should support:

- Employee creation
- Profile update
- Transfer
- Promotion
- Department change
- Manager change
- Salary revision
- Probation confirmation
- Resignation
- Exit process
- Full and final settlement

Mobile App

Employees should be able to:

- View profile
- Update limited personal details
- Upload documents
- View employment details
- Download letters and payslips

Approval Workflow

Profile changes can require approval from:

- HR
- Manager
- Admin

Module 3: Attendance Management

Objective

To automate employee attendance using multiple verification methods.

The system should support:

- QR-based attendance
- RFID-based attendance
- Thumb impression/biometric attendance
- Face recognition attendance
- GPS/location-based attendance
- Geo-fencing
- Manual attendance regularization
- Shift-based attendance
- Overtime calculation
- Late mark and early exit rules

Keka's attendance product positioning includes automated attendance, leave/absence tracking, shift/overtime integration, and payroll sync, which should be considered as benchmark-level functionality. (Keka HR)

3.1 Attendance Modes

A. QR-Based Attendance

The system should allow:

- Static QR for office/location
- Dynamic QR refreshed every few seconds/minutes
- Employee scans QR from mobile app
- System captures employee ID, date, time, device ID, GPS, and IP
- Anti-fraud validation using location and device binding

B. RFID-Based Attendance

The system should allow:

- RFID card assignment to employee
- RFID machine integration
- Check-in/check-out logs from RFID device
- Sync attendance logs to backend
- Retry mechanism if device is offline

C. Thumb Impression/Biometric Attendance

The system should allow:

- Employee biometric enrollment
- Device mapping with branch/location
- Biometric punch sync
- Duplicate punch prevention
- Device health monitoring

D. Face Recognition Attendance

The system should allow:

- Employee face enrollment
- Liveness detection
- Face match verification
- Geo-location validation
- Anti-spoofing checks

E. Location-Based Attendance

The system should allow:

- GPS-based punch-in/punch-out

- Geo-fencing for office/site
- Field staff attendance
- Work-from-home attendance
- Location accuracy threshold
- Distance validation from assigned location

3.2 Attendance Rules

The admin should be able to configure:

- Shift start and end time
- Grace period
- Half-day rules
- Late mark rules
- Early exit rules
- Minimum working hours
- Break time
- Overtime rules
- Weekly off
- Holiday rules
- Attendance regularization rules
- Auto absent marking
- Work from home policy
- Remote attendance policy

3.3 Attendance Workflows

Check-In Flow

1. Employee opens mobile app.
2. Selects attendance method.
3. System verifies identity.
4. System captures timestamp and location/device data.
5. Attendance is recorded.
6. Employee receives confirmation.

Check-Out Flow

1. Employee checks out using allowed method.
2. System calculates working hours.
3. System validates shift rules.
4. Attendance status is updated.

Regularization Flow

1. Employee submits missed punch/incorrect attendance request.
2. Manager reviews.
3. HR optionally reviews.
4. Attendance is corrected after approval.

Module 4: Leave Management

Objective

To automate employee leave application, approval, balance tracking, and payroll impact.

Functional Requirements

The system should support:

- Leave type configuration
- Casual leave
- Sick leave
- Earned leave
- Maternity leave
- Paternity leave
- Comp off
- Leave without pay
- Optional holidays
- Leave encashment

- Carry forward
- Leave accrual
- Sandwich leave rules
- Half-day leave
- Multi-level approval
- Leave cancellation
- Leave balance adjustment

Mobile App

Employee should be able to:

- Apply leave
- View leave balance
- Cancel leave
- Track approval status
- Upload medical/supporting documents

Manager App/Web

Manager should be able to:

- Approve/reject leave
- View team calendar
- View overlapping leaves
- Add remarks

Payroll Impact

Approved LWP should automatically flow into payroll.

Module 5: Shift, Roster & Overtime Management

Objective

To manage shift-based workforce scheduling.

Functional Requirements

The system should allow:

- Create shifts
- Assign shifts to employees
- Bulk roster upload
- Rotational shifts
- Night shifts
- Split shifts
- Weekly off planning
- Shift swap request
- Overtime request
- Auto overtime calculation
- Manager approval for overtime

Reports

- Shift-wise attendance
- Overtime summary
- Late coming report
- Early going report
- Absent report
- Roster compliance report

Module 6: Payroll Management

Objective

To automate salary processing, statutory deductions, payslip generation, and payroll reporting.

Keka's payroll references include automated payroll calculations, importing employee data, syncing attendance/reimbursements/loans, and statutory calculations for PF, ESIC, TDS, and PT. (Keka HR)

Functional Requirements

Payroll Configuration

The admin should be able to configure:

- Salary components
- Earnings
- Deductions
- Benefits
- Reimbursements
- Bonuses
- Incentives
- Arrears
- Loans and advances
- Gratuity
- Leave encashment
- Overtime payment
- Variable pay
- CTC structure
- Monthly salary structure
- Formula-based components

Salary Components

Examples:

- Basic
- HRA
- Conveyance
- Special allowance
- Medical allowance

- Bonus
- Incentive
- Employer PF
- Employee PF
- ESI
- Professional tax
- TDS
- LOP deduction
- Loan deduction

Payroll Process

The payroll process should include:

1. Employee data validation
2. Attendance and leave sync
3. Salary input validation
4. Earnings and deductions calculation
5. Statutory calculation
6. Tax calculation
7. Payroll preview
8. Approval workflow
9. Payslip generation
10. Bank statement generation
11. Payroll lock
12. Compliance report generation

Statutory Compliance

The system should support:

- PF
- ESIC
- Professional Tax
- TDS
- LWF, if applicable
- Form 16
- Payslip
- Salary register
- PF ECR report
- ESIC report

- PT report
- TDS report

For Indian payroll, EPFO references state that both employee and employer generally contribute 12% of basic wages plus dearness allowance, while ESIC currently lists employee contribution as 0.75% and employer contribution as 3.25% of wages. These rules should be implemented through a configurable statutory rule engine, not hardcoded. (EPFO)

Payroll Approval

Payroll should support:

- Draft payroll
- Payroll preview
- HR approval
- Finance approval
- Final payroll lock
- Payslip release

Payroll Reports

- Monthly payroll report
- Employee salary register
- Bank transfer statement
- Statutory deduction report
- Department-wise salary cost
- Location-wise payroll cost
- CTC report
- Variance report
- Full and final settlement report

Module 7: Expense & Reimbursement Management

Objective

To allow employees to submit expenses and enable organizations to approve, reimburse, and audit claims.

Functional Requirements

The system should support:

- Expense category configuration
- Travel claims
- Food claims
- Fuel claims
- Mobile/internet claims
- Medical claims
- Other reimbursements
- Bill upload
- OCR-based bill reading, optional
- Policy validation
- Approval workflow
- Finance approval
- Reimbursement through payroll or separate payment

Mobile App

Employees should be able to:

- Upload bill photo
- Enter expense details
- Submit claim
- Track status

Admin Web

Admin should be able to:

- Configure limits
- Approve/reject claims

- Export claim reports
- Sync approved claims with payroll

Module 8: Recruitment / ATS

Objective

To manage hiring from job requisition to offer rollout and onboarding.

Keka's ATS page mentions resume parsing, interview scheduling, candidate communication, AI screening, and candidate engagement as part of applicant tracking functionality. (Keka HR)

Functional Requirements

Job Requisition

The system should allow:

- Create hiring request
- Define position
- Department
- Location
- Experience range
- Salary range
- Job description
- Skills required
- Number of openings
- Approval workflow

Job Posting

The system should support:

- Internal job posting
- Career page
- Manual candidate upload
- Job board integration, future scope
- Referral candidate submission

Candidate Management

The system should support:

- Resume upload
- Candidate profile creation
- Resume parsing
- Candidate source tracking
- Application status tracking
- Duplicate candidate detection
- Candidate communication
- Interview scheduling
- Offer management

Hiring Pipeline

Example stages:

1. Applied
2. Resume screened
3. AI matched
4. Shortlisted
5. AI interview assigned
6. AI interview completed
7. Technical interview
8. HR interview
9. Offer released
10. Offer accepted
11. Onboarding initiated
12. Joined
13. Rejected/on hold

Module 9: AI-Based Resume Matching & Interview System

Objective

To automate candidate screening by comparing job descriptions with resumes and conducting AI-based interviews.

This is one of the key differentiators of the proposed HRMS.

9.1 Resume and JD Matching

Functional Requirements

The system should:

- Read job description
- Read candidate resume
- Extract skills
- Extract experience
- Extract education
- Extract certifications
- Extract project history
- Extract domain experience
- Compare resume with JD
- Generate match percentage
- Identify missing skills
- Identify strong areas
- Recommend candidate suitability
- Generate recruiter summary

AI Match Score Parameters

Parameter	Weight Example
Required skills match	30%
Relevant experience	20%
Domain exposure	15%
Education/certification	10%
Project relevance	15%
Stability/career progression	5%
Communication indicators	5%

Weights should be configurable by job role.

9.2 AI Interview Question Generation

The system should generate questions based on:

- Job description
- Resume
- Experience level
- Required skills
- Candidate claimed skills
- Role type
- Difficulty level
- Interview round

Question types:

- Technical questions
- Scenario-based questions
- Behavioral questions
- Communication questions
- Domain-specific questions
- Problem-solving questions
- Follow-up questions

9.3 AI Interview Execution

Candidate Flow

1. Candidate logs in through secure link.
2. Candidate verifies identity.
3. System checks camera/microphone permission.
4. AI interviewer starts interview.
5. Candidate answers by voice.
6. System records audio/video, if enabled.
7. Speech-to-text converts answer.
8. AI evaluates answer.
9. AI may ask follow-up questions.
10. Interview ends.
11. Report is generated.

Interview Modes

- Text-based AI interview
- Voice-based AI interview
- Video + voice interview
- MCQ assessment
- Coding assessment, future scope
- Proctored interview, future scope

9.4 AI Interview Scoring

The AI should score candidate answers on:

- Correctness
- Conceptual clarity
- Depth of knowledge
- Relevance to question
- Practical experience
- Communication
- Confidence

- Completeness
- Red flags
- Overall fitment

Output Report

The system should generate:

- Candidate summary
- Resume-JD match score
- Interview score
- Skill-wise score
- Question-wise answer evaluation
- Strengths
- Weaknesses
- Recommended next step
- Hiring risk
- Interview transcript
- Audio/video reference, if enabled

9.5 AI Guardrails

The system should include:

- Bias reduction prompts
- No discrimination based on gender, religion, caste, age, etc.
- Human review before rejection
- AI score explanation
- Audit trail of AI decisions
- Recruiter override option
- Candidate consent for AI interview

Module 10: Onboarding Management

Objective

To digitize the joining process after offer acceptance.

Functional Requirements

The system should support:

- Offer letter generation
- Offer approval
- Candidate acceptance
- Document collection
- Background verification tracking
- Joining form
- Asset request
- Email/account creation request
- Policy acknowledgment
- Digital employee profile creation
- Probation setup

Documents

- PAN
- Aadhaar
- Passport
- Address proof
- Education certificates
- Experience letters
- Bank proof
- Photograph
- Signed offer letter
- NDA
- Company policies

Module 11: Performance Management

Objective

To manage employee goals, OKRs, appraisals, feedback, and performance reviews.

Keka's product references include performance and culture, SMART goals, OKRs, feedback, and appraisal-related capabilities. (Keka HR)

Functional Requirements

The system should support:

- Goal setting
- OKR management
- KPI management
- Manager review
- Self-review
- 360-degree feedback
- Continuous feedback
- Quarterly review
- Annual appraisal
- Rating calibration
- Promotion recommendation
- Increment recommendation
- Performance improvement plan

Appraisal Workflow

1. HR creates appraisal cycle.
2. Employees submit self-review.
3. Managers submit review.
4. Reviewer/department head approves.
5. HR calibrates ratings.
6. Final rating is published.
7. Increment/promotion is processed.

Module 12: Employee Engagement & Culture

Objective

To measure and improve employee engagement.

Functional Requirements

The system should support:

- Employee surveys
- Pulse surveys
- Anonymous feedback
- Mood tracker
- Recognition and rewards
- Announcements
- Birthday/work anniversary notifications
- Company newsfeed
- Polls
- Engagement analytics

Module 13: Employee Self-Service

Objective

To reduce dependency on HR by allowing employees to access common HR services themselves.

Functional Requirements

Employees should be able to:

- View profile
- Apply leave
- Mark attendance
- View attendance
- View payslips
- Download Form 16
- Submit expenses
- Raise helpdesk tickets
- View company policies
- Upload documents
- Submit tax declarations
- View team directory
- Update personal details
- View holidays
- View shifts
- Check approval status

Keka's HRMS positioning also highlights employee self-service as a core HRMS capability. (Keka HR)

Module 14: Manager Self-Service

Objective

To allow managers to manage team-level approvals and workforce visibility.

Functional Requirements

Managers should be able to:

- View team attendance
- Approve/reject leaves
- Approve/reject regularization
- Approve expenses
- View team calendar

- Conduct performance reviews
- Assign goals
- Review timesheets
- Raise hiring requests
- View team analytics

Module 15: Helpdesk / Ticketing

Objective

To manage HR, IT, payroll, admin, and employee queries.

Functional Requirements

The system should support:

- Ticket creation
- Ticket category
- Priority
- Assignment
- SLA configuration
- Comments
- Attachments
- Escalation
- Closure
- Reopen
- Employee feedback

Ticket categories:

- HR
- Payroll
- IT
- Attendance
- Leave

- Asset
- Finance
- Admin

Module 16: Project Timesheet & PSA

Objective

To track project work, billable hours, resource utilization, and project profitability.

Keka's product references include timesheets, projects, resource management, project profitability, and PSA functionality. (Keka HR)

Functional Requirements

The system should support:

- Project creation
- Client creation
- Task creation
- Employee allocation
- Timesheet submission
- Billable/non-billable hours
- Approval workflow
- Project cost tracking
- Resource utilization
- Milestone tracking
- Invoice data generation

Reports

- Employee timesheet report
- Project profitability report
- Billable utilization report

- Client-wise hours report
- Department-wise utilization report

Module 17: Asset Management

Objective

To manage company assets issued to employees.

Functional Requirements

The system should support:

- Asset master
- Asset category
- Asset assignment
- Asset return
- Asset condition
- Warranty details
- Asset transfer
- Exit clearance integration

Asset examples:

- Laptop
- Mobile
- ID card
- RFID card
- Access card
- SIM card
- Tools/equipment

Module 18: Document & Letter Management

Objective

To manage employee documents and HR letters.

Functional Requirements

The system should support:

- Document upload
- Document verification
- Document expiry alerts
- Letter templates
- Offer letter
- Appointment letter
- Confirmation letter
- Increment letter
- Promotion letter
- Experience letter
- Relieving letter
- Warning letter
- NDA
- Policy acknowledgment

Module 19: Exit Management

Objective

To manage resignation, clearance, and full and final settlement.

Functional Requirements

The system should support:

- Resignation submission
- Manager approval
- HR approval
- Notice period calculation
- Exit interview
- Asset clearance
- IT clearance
- Finance clearance
- Leave encashment
- Recovery calculation
- Full and final settlement
- Relieving letter
- Experience letter

Module 20: Analytics & Reporting

Objective

To provide real-time HR and workforce insights.

Dashboard Requirements

HR Dashboard

- Total employees
- New joiners
- Exits
- Attrition rate
- Department-wise headcount
- Gender diversity
- Attendance summary
- Leave summary
- Hiring status

- Payroll cost

Payroll Dashboard

- Monthly payroll cost
- Department-wise salary cost
- Statutory deductions
- Payroll variance
- Pending payroll inputs
- LOP summary

Attendance Dashboard

- Present
- Absent
- Late
- Early exit
- Work from home
- Field attendance
- Overtime

Recruitment Dashboard

- Open positions
- Candidates applied
- Shortlisted candidates
- AI match score distribution
- Interview pending
- Offer released
- Offer accepted
- Time to hire

Performance Dashboard

- Goal completion
- Rating distribution
- High performers
- Low performers
- Department-wise performance

Module 21: Notifications & Communication

Objective

To keep users informed through automated alerts.

Channels

- In-app notification
- Email
- SMS
- WhatsApp, optional
- Push notification

Notification Events

- Leave approval
- Attendance regularization
- Payroll release
- Payslip generated
- Expense approval
- Interview scheduled
- AI interview link
- Offer released
- Birthday/work anniversary
- Policy update
- Ticket update
- Appraisal reminder

Module 22: Admin Configuration

Objective

To allow no-code/low-code configuration of HR policies.

Configurable Masters

- Departments
- Designations
- Branches
- Grades
- Leave types
- Holiday calendar
- Shift rules
- Attendance rules
- Payroll components
- Salary templates
- Approval workflows
- Expense policies
- Recruitment stages
- Interview templates
- Performance cycles
- Ticket categories
- Document types
- Letter templates
- Roles and permissions

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6. Backend Architecture

Recommended Architecture

For the first version, use a **modular monolith backend** with clean module boundaries. Later, split into microservices when usage grows.

Suggested backend modules:

1. Identity & Access Module
2. Organization Module
3. Employee Module
4. Attendance Module
5. Leave Module
6. Payroll Module
7. Expense Module
8. Recruitment Module
9. AI Interview Module
10. Performance Module
11. Timesheet Module
12. Helpdesk Module
13. Notification Module
14. Reporting Module
15. Integration Module
16. Audit & Compliance Module

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7. Recommended Tech Stack

Option A: Practical Stack for Faster Development

Layer	Technology
Admin Web App	Next.js + TypeScript
UI Library	Ant Design / Tailwind CSS
Mobile App	React Native
Backend API	Django REST Framework / FastAPI
AI Service	Python FastAPI microservice
Database	PostgreSQL
Cache	Redis
Background Jobs	Celery + Redis / RabbitMQ
File Storage	AWS S3 / Azure Blob
Authentication	JWT + Refresh Token + MFA
Search	PostgreSQL Full Text / OpenSearch later
Reports	Python Pandas + Excel/PDF generation
Notifications	Firebase Cloud Messaging, Email SMTP, SMS Gateway

Deployment	Docker + Nginx
Cloud	AWS / Azure / GCP
CI/CD	GitHub Actions
Monitoring	Sentry + Prometheus/Grafana
Logs	ELK / OpenSearch / CloudWatch

Option B: Enterprise Microservices Stack

Layer	Technology
Web Frontend	Next.js
Mobile	Flutter / React Native
API Gateway	Kong / Nginx / AWS API Gateway
Backend Services	NestJS / Django / FastAPI
Database	PostgreSQL per service or shared schema initially
Event Bus	Kafka / RabbitMQ
Cache	Redis
AI/ML Service	Python FastAPI
Object Storage	S3
Authentication	Keycloak / Auth0 / Custom IAM
Analytics	Metabase / Superset / Custom BI
Deployment	Kubernetes
Observability	Prometheus, Grafana, Loki, Sentry

My Recommended Stack for You

Considering your existing work with Django, PostgreSQL, Next.js, APIs, dashboards, and role-based systems:

- **Admin Web App:** Next.js + TypeScript + Tailwind/Ant Design
- **Mobile App:** React Native
- **Backend:** Django REST Framework
- **AI Interview Service:** Python FastAPI
- **Database:** PostgreSQL
- **Async Jobs:** Celery + Redis
- **File Storage:** AWS S3

- **Deployment:** Docker + Nginx
- **Cloud:** AWS
- **Future Scaling:** Break payroll, attendance, recruitment, and AI interview into independent services later

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8. AI/ML Tech Stack

Requirement	Suggested Tool
Resume parsing	Python, spaCy, PyMuPDF, docx parser
JD parsing	LLM-based extraction
Resume-JD matching	Embeddings + rule-based scoring
AI question generation	OpenAI / Azure OpenAI / open-source LLM
Speech-to-text	Whisper / Azure Speech / Google Speech
Answer evaluation	LLM scoring + rubric
Candidate report	LLM + structured JSON output
Face verification	AWS Rekognition / Azure Face / custom ML
Liveness detection	Mobile SDK / third-party biometric SDK
Document OCR	Tesseract / AWS Textract / Azure OCR

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9. High-Level System Architecture

```
Employee / Candidate Mobile App
        |
        | REST / WebSocket / Push
        |
Admin Web App / Manager Portal
        |
        v
API Gateway / Backend API
        |
        |--------------- Identity & Access
        |--------------- Employee Management
        |--------------- Attendance
```

```
        |--------------- Leave
        |--------------- Payroll
        |--------------- Recruitment
        |--------------- AI Interview
        |--------------- Performance
        |--------------- Expense
        |--------------- Timesheet
        |--------------- Helpdesk
        |--------------- Reports
        |
        v
PostgreSQL + Redis + S3
        |
        v
Background Workers / Notification Engine
        |
        v
Email / SMS / WhatsApp / Push Notifications
```

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10. Key Database Entities

Core entities should include:

- tenants
- users
- roles
- permissions
- employees
- employee_documents
- departments
- designations
- branches
- shifts
- attendance_logs
- attendance_regularization
- leave_types
- leave_balances
- leave_requests
- salary_components
- salary_structures
- payroll_runs

- payroll_items
- payslips
- statutory_rules
- expenses
- expense_items
- job_requisitions
- job_posts
- candidates
- resumes
- interview_rounds
- ai_interviews
- ai_interview_questions
- ai_interview_answers
- ai_interview_reports
- onboarding_tasks
- goals
- okrs
- appraisal_cycles
- performance_reviews
- projects
- timesheets
- helpdesk_tickets
- assets
- notifications
- audit_logs

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11. Major API Groups

Authentication APIs

- Login
- Logout
- Refresh token
- Forgot password
- MFA verification
- Role permissions

Employee APIs

- Create employee
- Update employee
- Get employee profile
- Upload documents
- Employee lifecycle actions

Attendance APIs

- QR check-in
- QR check-out
- GPS attendance
- Biometric sync
- RFID sync
- Attendance report
- Regularization request
- Approval action

Payroll APIs

- Salary structure
- Payroll preview
- Payroll process
- Payslip generation
- Payroll lock
- Statutory reports

Recruitment APIs

- Create job
- Apply candidate
- Upload resume
- Resume parsing
- JD matching
- Schedule interview
- Generate offer

AI Interview APIs

- Generate questions
- Start interview
- Submit answer
- Transcribe answer
- Evaluate answer
- Generate report

Leave APIs

- Apply leave
- Approve leave
- Reject leave
- Leave balance
- Leave calendar

Expense APIs

- Submit expense
- Upload bill
- Approve expense
- Reject expense
- Sync with payroll

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12. Security Requirements

The system should include:

- Role-based access control
- Multi-factor authentication
- JWT/secure session management
- Password policy
- Data encryption at rest
- Data encryption in transit
- Audit logs
- IP restriction, optional
- Device binding for mobile attendance
- Biometric data protection

- Consent for AI interviews
- Consent for face/location attendance
- Payroll data masking
- Secure file upload
- Virus scanning for uploaded files
- API rate limiting
- Admin activity tracking

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13. Compliance Requirements

The platform should support Indian payroll and HR compliance, including:

- PF
- ESIC
- TDS
- Professional Tax
- LWF, where applicable
- Payslip generation
- Salary register
- Form 16 support
- Audit trail
- Statutory report export

Because payroll laws and rates can change, statutory values should be stored in a **configurable compliance rule engine** rather than hardcoded.

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14. Non-Functional Requirements

Area	Requirement
Performance	Common screens should load within 2–3 seconds
Scalability	System should support multi-tenant scaling
Availability	Target 99.5%+ uptime initially
Security	All sensitive data encrypted

Auditability	All critical changes logged
Mobile Support	Android and iOS support required
Offline Handling	Attendance device sync should support offline retry
Reporting	Excel/PDF export required
Backup	Daily database backup
Disaster Recovery	Backup restore process required
Localization	India-first; multi-language optional
Extensibility	Payroll, attendance, and AI modules should be configurable

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15. Suggested Development Phases

Phase 1: MVP Core HRMS

Recommended duration: 10–14 weeks

Scope:

- Login and roles
- Organization setup
- Employee management
- Attendance basic: QR + GPS
- Leave management
- Basic payroll
- Payslip
- Admin dashboard
- Employee mobile app basic
- Notifications

Phase 2: Advanced HR Operations

Recommended duration: 10–12 weeks

Scope:

- RFID/biometric integration
- Shift/roster
- Overtime
- Expense management
- Helpdesk
- Document management
- Onboarding
- Reports and analytics

Phase 3: Recruitment & AI Interview

Recommended duration: 8–12 weeks

Scope:

- ATS
- Resume parsing
- JD matching
- AI question generation
- Voice interview
- Answer scoring
- Candidate report
- Offer management

Phase 4: Performance, PSA & Enterprise Features

Recommended duration: 10–12 weeks

Scope:

- OKR
- Appraisal
- 360 feedback
- Engagement surveys
- Project timesheet
- Resource allocation
- Billing reports

- Advanced analytics

Phase 5: Enterprise Scaling

Scope:

- Microservices split
- Advanced compliance engine
- BI dashboards
- HR chatbot
- Predictive analytics
- Attrition prediction
- Workforce planning
- Third-party integrations

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16. What More Can Be Included

Apart from Keka-like features, you can add these differentiators:

AI HR Assistant

An internal chatbot for:

- Leave policy questions
- Payroll FAQs
- Attendance queries
- HR ticket creation
- Payslip explanation
- Candidate screening summary

Attrition Prediction

AI can predict employees at risk of leaving based on:

- Attendance behavior
- Performance trend
- Engagement score
- Manager feedback
- Salary revision history

Workforce Planning

Management can forecast:

- Hiring needs
- Department capacity
- Skill gaps
- Project resource availability

Compliance Risk Dashboard

Shows:

- Missing employee documents
- Pending statutory data
- Payroll errors
- Attendance anomalies
- Policy violations

Smart Attendance Fraud Detection

Detects:

- Same device used by multiple employees
- GPS spoofing possibility
- Repeated regularization abuse
- QR scan outside allowed area
- Face mismatch

AI-Based Job Description Builder

HR enters role name and system creates:

- JD
- Required skills
- Interview questions
- Evaluation rubric
- Salary benchmark, optional

Employee Sentiment Analysis

Using surveys and feedback to understand:

- Employee morale
- Department-level issues
- Manager feedback trend
- Engagement risk

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17. Key Assumptions

- The system will initially be India-focused.
- Payroll compliance will be configurable as rules may change.
- Biometric/RFID attendance will depend on hardware device APIs.
- AI interview decisions will be recommendation-based, not final automatic rejection.
- Candidate consent will be taken before AI interview recording and evaluation.
- Admin web app and mobile app will share the same backend APIs.
- Multi-tenant architecture should be planned from the beginning.

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18. Out of Scope for Initial MVP

These can be future scope:

- Global payroll

- Complex union/labor contract management
- Advanced workforce budgeting
- Native biometric device manufacturing
- Full proctored coding assessment
- Background verification vendor integration
- ERP/accounting integration
- WhatsApp bot
- HR voice bot
- Predictive attrition model

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19. Acceptance Criteria

The system will be considered successful when:

- Admin can create company, employees, roles, policies, and salary structures.
- Employees can mark attendance from mobile app.
- Attendance flows into leave/payroll calculation.
- Leave approval works end-to-end.
- Payroll can be processed with earnings, deductions, statutory components, and payslips.
- Candidates can upload resumes and apply for jobs.
- AI can compare JD and resume and generate match score.
- AI interview can ask questions, record answers, score responses, and generate report.
- Managers can approve requests.
- HR can generate reports.
- Admin can configure policies without developer dependency.
- All critical actions are audit logged.

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20. Final Recommended MVP Scope

For the first production-ready release, I would not build everything at once. I would start with:

1. Core HR and employee master
2. Role-based admin panel
3. Android/iOS employee app
4. QR + GPS attendance

5. Leave management
6. Shift and holiday management
7. Basic payroll and payslip
8. Recruitment ATS basic
9. AI resume-JD matching
10. AI interview prototype
11. Reports and dashboards
12. Notifications
13. Audit logs and security

Then expand into RFID, biometric, full payroll compliance, performance, PSA, helpdesk, and advanced AI.

This way the platform becomes usable early, while the enterprise modules can be added in controlled phases.